

Data Analytics Capstone Project

During the last 4 weeks of the Data Analytics Bootcamp, students work full-time on their own capstone project to put the concepts learned in the bootcamp into practice, deepening knowledge and strengthening abilities, as well as providing an example of work for their portfolio.

The capstone process is outlined here to give you a better understanding of the contents and requirements of the capstone project.

Capstone in a nutshell:

In the capstone project, students should, in the end, create a 10-15 minute presentation that answers a specific question about a business topic. The project involves finding the topic and iteratively developing and clarifying the question, along with finding the data and performing the analytical steps required to answer the question. Students tend to work with between 2 and 4 separate data sources.

Required deliverables:

- Clearly stated business question with the background and impact explained.
- Technical data analysis completed using Python and SQL, plus additionally what suits your needs - such as Tableau, spreadsheets, ...
- Work is stored in a GitHub repository.
- Data is stored in an SQL database (Neuefische can provide it during project time).
- Final presentation of 10-15 minutes, accompanying slides are included in the GitHub repo.

Data reuse is granted, though your project must be genuinely yours and distinguishable from projects done before. tl:dr - no copy/paste!

Capstone project outcome effects

Group size: The capstone project can be done individually or in a group. Group sizes of 3 or 4 members are recommended.

Graduation Event: Small group sizes may reduce visibility during the final graduation event because more presentations need to be held.

Certificates: Not fulfilling the required deliverables may result in the bootcamp ending with just a certificate of attendance instead of a Data Analyst certification.

Capstone and data analytics workflow steps:

Project definition: Understand stakeholder needs, potential outcomes, actions, areas of uncertainty, and success criteria, plus measurement of success.

Scope out project in reverse order: Desired outcome, potential actions, decisions to be made, predictions to be used, data requirements.

Find data: Access data via provided files, databases, and/or using APIs.

Clean and manipulate data: Data wrangling using taught techniques such as identifying duplicates, finding missing values, and detecting outliers. Store data in an SQL database.

Understand data: Exploratory data analysis to establish the validity of the dataset. Finding the distribution of data, the presence of outliers, and the correlation between different columns.

Generate insights: Data visualization using Python libraries and/or Tableau to highlight the distribution of data, relationships between data, and the composition and comparison of data.

Communicate insights to stakeholders: Charts and dashboard construction using Tableau or Python to highlight insights in an intuitive, appealing, and easy-to-understand way.

Drive decision-making and action: Presentation demonstrating your ability to clarify your analysis and relay your findings in a way that is easy to understand.